

Trends in catheter-directed therapy and in-hospital outcomes among patients with acute pulmonary embolism: insights from a multicentre national quality assurance database registry

Ioannis T. Farmakis ^{1,2}, Steven Horbal ³, John M. Moriarty ⁴, Mahir Elder⁵, Thomas Todoran ⁶, Rachel P. Rosovsky ⁷, Eric Lehr ⁸, Matthew D. Langston ⁹, Seth I. Sokol ¹⁰, Kenneth Rosenfield ¹¹, Robert Lookstein¹², Eric Secemsky ¹³, Konstantinos C. Christodoulou ¹, Lukas Hobohm ¹, Luca Valerio ¹, Stefano Barco ^{1,14}, and Stavros V. Konstantinides ^{1,15*}

¹Center for Thrombosis and Hemostasis, University Medical Center of the Johannes Gutenberg University, Langenbeckstrasse 1, Mainz 55131, Germany; ²Medizinische Klinik und Poliklinik I, LMU Klinikum, LMU München, Munich, Germany; ³Michigan Clinical Outcomes Research and Reporting Program, University of Michigan, Ann Arbor, MI, USA; ⁴David Geffen School of Medicine at UCLA, Division of Interventional Radiology, UCLA Medical Center, Los Angeles, CA, USA; ⁵School of Medicine, Wayne State University, Detroit, MI, USA; ⁶Medical University of South Carolina Division of Cardiology, Charleston, SC, USA; ⁷Harvard Medical School; Division of Hematology and Oncology, Department of Medicine, Massachusetts General Hospital, Boston, MA, USA; ⁸Division of Cardiac Surgery, Swedish Heart and Vascular Institute, Seattle, Washington, USA; ⁹Albert Einstein College of Medicine, Department of Medicine, Division of Pulmonology and Critical Care, Bronx, New York, USA; ¹⁰Albert Einstein College of Medicine, Department of Medicine, Division of Cardiology, Bronx, New York, USA; ¹¹Harvard University School of Medicine Division of Cardiology, Boston, MA, USA; ¹²Mount Sinai School of Medicine Division of Interventional Radiology, New York, New York, USA; ¹³Harvard University School of Medicine Division of Cardiology, Beth Israel Deaconess and Leahy Clinic Medical Center, Boston, MA, USA; ¹⁴Department of Angiology, University Hospital Zurich, Zurich, Switzerland; and ¹⁵Department of Cardiology, Democritus University of Thrace, Alexandroupolis 69100, Greece

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Aims

Multidisciplinary pulmonary embolism response teams (PERTs) are being established in hospitals worldwide to address the increasing complexity in acute PE management. To identify recent trends in PERT decisions regarding advanced treatment of acute severe PE.

Methods and results

We analysed data from the prospective multicentre PERTTM Consortium registry (years 2018–2024), focusing on catheter-directed treatment (CDT) and including systemic thrombolysis, surgical embolectomy, and extracorporeal membrane oxygenation (ECMO). An age-, sex-, and PE risk-matched population from the US Nationwide Inpatient Sample (NIS) was used for comparison. Among 11 436 patients enrolled at 51 sites (median age, 65 years; 13.7% high-risk and 62.5% intermediate-risk PE), 2639 (23.1%) underwent CDT. Of those, 140 (5.3%) underwent catheter-directed thrombolysis without ultrasound, 851 (32.2%) ultrasound-assisted catheter thrombolysis, and 1534 (58.1%) mechanical thrombectomy/aspiration. Systemic thrombolysis was used in 5.6%, surgical embolectomy in 1.1%, and ECMO in 1.6% of all patients. Trends of CDT increased over time (+0.36% quarterly by linear regression; $P = 0.002$), with increase in mechanical thrombectomy (+0.83%; $P < 0.001$) and decrease in catheter-directed thrombolysis (−0.4%; $P = 0.001$). Matching 10 883 patients from the PERTTM Consortium registry to the NIS population, we found a 22% (95% CI, 21–23%) standardized mean difference in CDT use, 1.3% (0.6–2.0%) lower in-hospital mortality, and 0.75 (0.2–1.3) fewer days of hospital stay among PERTTM Consortium registry patients.

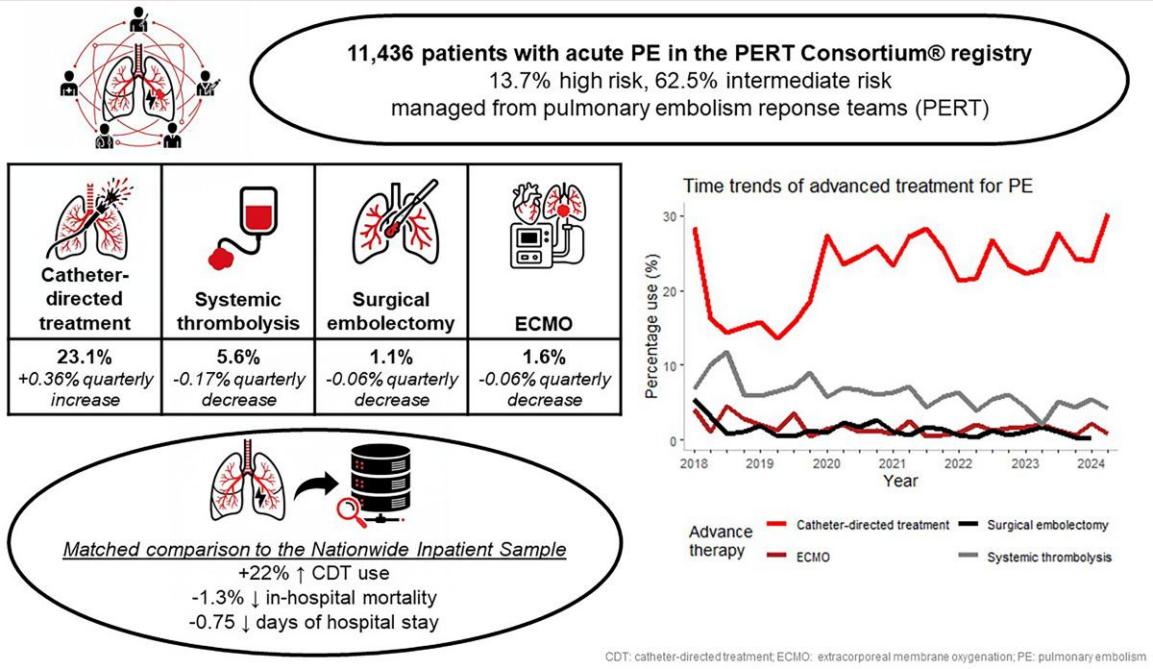
* Corresponding author. Tel: 06131 17 8382, Email: stavros.konstantinides@unimedizin-mainz.de

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Conclusion

In a national quality assurance database of patients with PE included in the PERT registry, the use of catheter-directed treatment increased over time. Compared with a nationwide NIS sample, these patients had lower in-hospital mortality and shorter hospital length of stay.

Graphical abstract



Keywords

pulmonary embolism • Pulmonary embolism response team • Advanced therapies • Catheter directed treatment • Systemic thrombolysis

Introduction

Acute pulmonary embolism (PE) is a potentially life-threatening acute cardiovascular event imposing a substantial burden to patients, their families, and the society.^{1,2} Although recent epidemiological data suggest an overall decrease in PE-related mortality since the beginning of this century,^{3,4} certain population groups remain at persistently elevated and even increasing risk for death over time.⁴⁻⁶ Traditionally, systemic thrombolysis has been recommended as first-line reperfusion treatment in patients presenting with haemodynamic compromise due to PE or those at risk of imminent decompensation.⁷⁻⁹ More recently, however, catheter-directed treatment (CDT), encompassing various modalities such as catheter-directed thrombolysis with or without ultrasound assistance, large-bore mechanical thrombectomy, and computer-assisted vacuum thrombectomy, emerged as an alternative for managing patients with intermediate-high-risk or high-risk PE.^{10,11} While awaiting the results of major randomized controlled trials (RCTs) comparing CDT with the current standard of care and focusing on clinically relevant endpoints,¹² smaller trials with surrogate endpoints, single-arm cohort studies, and a number of registries have yielded promising results regarding both the efficacy and the safety of interventional systems and procedures.¹³ In addition to technical innovations, the concept of PE response teams (PERT) addressing the need

for a multidisciplinary approach to the increasing complexity in the management of acute severe PE, is progressively being integrated into the state-of-the-art PE care in hospitals in the US and across the world.^{14,15} PERT implementation helps to personalize treatment of patients with acute high-risk or intermediate-risk PE, making the best use of the local expertise and resources, and it may contribute to an improved in-hospital outcome, including higher chances of survival in the acute phase.^{16,17}

Against this background, we sought to examine recent trends in PERT decision-making, the utilization of advanced treatment options, and outcomes in patients with acute PE. For this purpose, we analysed a multicentre national quality assurance database registry of centres dedicated to providing multidisciplinary care for PE patients and compared the results to those obtained from a nationwide cohort.

Methods

Patient population

We analysed data from the PERT™ Consortium national quality assurance database registry, collecting information on consecutive PERT activations and consultations at participating centres across the US.¹⁸⁻²⁰ The registry is designed to examine national trends in clinical practice and patient

outcomes, while also providing benchmarking tools to support continuous quality improvement at each site. The present analysis includes data from patients admitted to a total of 51 participating US sites between January 2018 and May 2024. Patient data was anonymized and securely stored in the REDCap (Research Electronic Data Capture) system, with centralized dataset management handled by the Michigan Clinical Outcomes Research & Reporting Program at the University of Michigan. Institutional Review Board (IRB) approval was obtained either locally at participating sites or through a central IRB (Advarra).

For the control population, we accessed the US Nationwide Inpatient Sample (NIS), developed by the Agency for Healthcare Research and Quality for the Healthcare Cost and Utilization Project (HCUP). The NIS is the largest publicly available, all-payer, de-identified inpatient healthcare database in the United States. Representing approximately 97% of the U.S. population, it includes a 20% stratified sample of discharges from hospitals participating in HCUP, excluding rehabilitation and long-term acute care facilities. For this study, we analysed data from the NIS spanning 2016–2020. As the database contains only de-identified information, data were exempt from institutional review board approval and informed consent requirements. Further details about the NIS are available on the HCUP NIS Database Documentation website (www.hcup-us.ahrq.gov).

Definitions of baseline parameters and outcomes

We analysed all consecutive patients with a diagnosis of acute PE included in the PERT™ Consortium registry. Baseline characteristics (including demographics, comorbidities, and risk factors for PE), as well as hospitalizations (including diagnostic modalities, treatment procedures, and clinical outcomes) were analysed as recorded in the centralized dataset. Patients were classified by each participating site's PERT into three risk categories: Low-, intermediate-, and high-risk PE. For the purpose of the analysis, the risk stratification provided by the PERT of each site was used. However, patients for whom a PERT stratification was missing were re-evaluated and classified as follows. High-risk PE was defined as haemodynamic instability (systolic blood pressure < 90 mmHg over 15 min, signs and symptoms of shock, and/or need for vasopressor or inotropic support). Intermediate-risk PE was defined as haemodynamic stability plus evidence of right ventricular dysfunction in computed tomography (CT) or echocardiography, and/or elevated cardiac biomarkers. The remaining patients were classified into the low-risk PE category. Advanced therapy was defined as systemic thrombolysis, catheter-directed treatment (CDT), surgical embolectomy, and extracorporeal membrane oxygenation (ECMO). CDT was further categorized into standard catheter-directed thrombolysis, ultrasound-assisted catheter-directed thrombolysis, and catheter-directed thrombectomy.

Data definitions in the NIS database were performed as described before in similar analyses of patients with PE performed by our research group.^{21,22} [Supplementary material online, Table S1](#) lists the ICD-10-CM and ICD-10-PCS codes used to define the aforementioned variables in the NIS database.

The study's outcomes included the use of advanced therapies across years, in-hospital all-cause mortality, in-hospital major bleeding, and in-hospital length of stay. In-hospital major bleeding was defined based on the International Society on Thrombosis and Haemostasis (ISTH) criteria. This definition included fatal bleeding or bleeding occurring in critical areas such as intracranial, intraspinal, intraocular, retroperitoneal, intra-articular, or pericardial regions; intramuscular bleeding associated with compartment syndrome; or bleeding that resulted in a haemoglobin drop of 2 g/dL or required the transfusion of at least two units of whole blood or red blood cells.

Statistical analysis

The cleaned dataset of the PERT™ Consortium registry was provided by the Michigan Clinical Outcomes Research & Reporting Program at the University of Michigan; the analysis was performed at the Centre for Thrombosis and Haemostasis at the University Medical Centre Mainz. A complete case analysis was performed. Descriptive statistics for categorical data were presented as counts and percentages, while continuous variables were summarized using medians and interquartile ranges (IQRs). Trends of advanced treatments over time in the PERT™ Consortium registry were

evaluated by fitting linear regression models. The linearity assumption was assessed through visual inspection of scatterplots and residual plots, and further tested by including a quadratic time term where appropriate, which supported the appropriateness of a linear fit.

We performed 1:1 matching of patients from the PERT™ Consortium registry and the NIS database. Given shifting practice over time, we performed a primary analysis including only patients from 2018 to 2020 to match the timeframe of inclusion across the two population sources. A secondary exploratory analysis was performed by including patients across the entire period available for both cohorts. For the matching procedure, we used greedy nearest neighbour propensity score matching without replacement. Propensity scores were generated using logistic regression, with the data source (PERT™ Consortium registry vs. NIS database) as the dependent variable and age, sex, and risk stratification as covariates to adjust for confounding. Covariate balance was evaluated using a Love plot, with standardized mean differences below 0.15 for all covariates signifying adequate balance. In the matched groups, the Welch two-sample t-test was used to compare in-hospital length of stay, while the two-sample test for equality of proportions was applied to compare in-hospital mortality and in-hospital major bleeding. A two-sided *P* value < 0.05 was set as the level of statistical significance.

The statistical analysis was conducted in R (R Project for Statistical Computing, version 4.1.3.).

Results

A total of 51 sites participating in the PERT™ Consortium registry enrolled 11 436 patients with a median age of 65 years (IQR 53–75; 49% women) between January 2018 and May 2024. A list of the participating institutions is provided in [Supplementary material online, Table S2](#). Most patients (7142; 62.5%) were classified into the intermediate-risk category, while 1569 (13.7%) had high-risk PE. For 2180 patients (19.1%), a PERT-reported risk classification was missing; after re-evaluation of the clinical parameters, there were only 183 patients (1.6%) whose recorded data were insufficient to determine the PE risk class. The baseline characteristics of included patients are provided in [Table 1](#).

Of all patients with acute PE, 2639 (23.1%) underwent CDT. Specifically, catheter-directed thrombolysis without ultrasound was employed in 140 (5.3% of all CDT cases), ultrasound-assisted catheter-directed thrombolysis in 851 (32.2%), and catheter-directed mechanical thrombectomy in 1534 (58.1%); data were missing for 114 patients. Systemic thrombolysis was used in 5.6% and surgical embolectomy in 1.1% of all patients. ECMO was required for haemodynamic support in 1.6%.

Among 1569 patients belonging to the high-risk PE class, CDT was performed in 574 (36.6%); of these, 27 (4.7%) underwent catheter-directed thrombolysis without ultrasound, 180 (31.3%) ultrasound-assisted catheter-directed thrombolysis, and 335 (58.4%) catheter-directed mechanical thrombectomy ([Figure 1](#)). Systemic thrombolysis was used in 19.2%, while surgical embolectomy was used in 3.3% of high-risk patients. ECMO was required for haemodynamic support in 7.6% of the cases ([Figure 1](#)).

Time trends in the use of advanced reperfusion therapy, and of specific CDT modalities across risk categories are depicted in [Figure 2](#) and [Figure 3](#) respectively. CDT use amounted to 17.0% in 2018 and 15.9% in 2019, with an abrupt increase to 25.3% in 2020, and stabilization thereafter, reaching 25.9% of PE cases in 2024 ([Figure 2](#)). The linear regression model calculated a 0.36% quarterly increase in CDT use (*P* = 0.002). Among patients treated with CDT, use of mechanical thrombectomy/aspiration increased during the study period ([Figure 3](#)), from 2.4% in 2018 to 20.0% in 2024 (quarterly increase, +0.83%, *P* < 0.001), this trend was paralleled by a decrease in catheter-directed thrombolysis from 12.8% in 2018 to 3.8% in 2024 (ultrasound-assisted, −0.4% per quarter, *P* < 0.001; standard, −0.1% per quarter, *P* = 0.007). Regarding systemic thrombolysis, usage rates reached 7.8% in 2018,

Table 1 Baseline characteristics of patients in the PERT™ consortium registry according to PE risk class

Characteristic	Overall n = 11 436 ^a	High risk n = 1 569 ^a	Intermediate risk n = 7 142 ^a	P value ^b
Age	65 (53, 75)	66 (55, 75)	66 (55, 76)	0.5
Sex				
Female	5574 (49%)	771 (49%)	3465 (49%)	
Male	5835 (51%)	796 (51%)	3660 (51%)	
Other	4 (<0.1%)	0 (0%)	2 (<0.1%)	
Body mass index	30 (26, 36)	31 (26, 36)	31 (26, 37)	0.9
Syncope	1138 (10.0%)	294 (19%)	733 (10%)	<0.001
Hypoxia	4175 (37%)	852 (54%)	2736 (38%)	<0.001
Active malignancy	1753 (15%)	270 (17%)	1084 (15%)	0.3
Deep vein thrombosis	2092 (18%)	259 (17%)	1318 (18%)	0.2
Right ventricular dysfunction	7478 (65%)	1266 (81%)	5257 (74%)	<0.001
Natriuretic peptide level elevation	5874 (73%)	894 (79%)	4170 (81%)	0.2
Troponin elevation	7073 (72%)	1175 (87%)	4900 (79%)	<0.001
Intensive care unit admission	5428 (47%)	1261 (80%)	3492 (49%)	<0.001
Family history of venous thromboembolism	855 (7.5%)	88 (5.6%)	540 (7.6%)	0.015
Smoking	1698 (15%)	179 (11%)	1054 (15%)	0.036
Recent trauma	657 (5.7%)	106 (6.8%)	386 (5.4%)	0.10
Hormonal treatment	659 (5.8%)	94 (6.0%)	405 (5.7%)	0.4
Reduced mobility	3494 (31%)	524 (33%)	2088 (29%)	0.022

^aMedian (IQR), or n (%).
^bCalculated by Kruskal–Wallis rank sum test, Pearson’s chi-squared test, or Fisher’s exact test as appropriate.

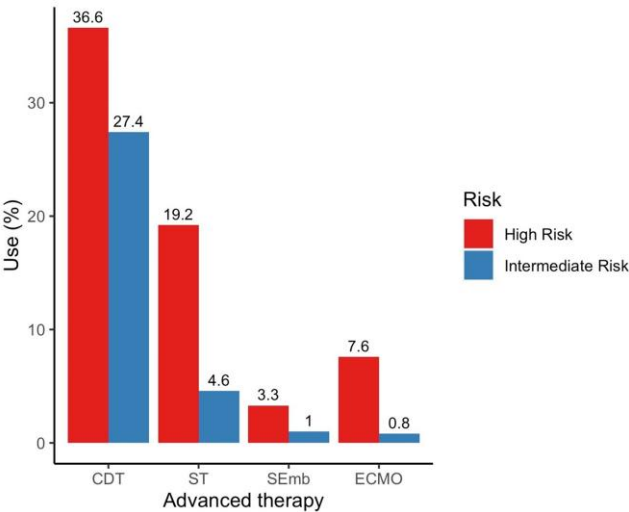


Figure 1 Advanced reperfusion treatment use in patients with high- and intermediate-risk pulmonary embolism in the PERT™ consortium registry, 2018–2024. CDT: catheter-directed treatment; ECMO: extracorporeal membrane oxygenation; SEmb: surgical embolectomy; ST: systemic thrombolysis.

with a decrease thereafter, reaching 3.9% and 5.0% in 2023 and 2024, respectively (−0.17% for every yearly quarter, $P < 0.001$; [Figure 2](#)). Surgical embolectomy (−0.06% for every yearly quarter, $P = 0.048$) and ECMO (−0.06% for every yearly quarter, $P = 0.02$) did not show

remarkable fluctuations in usage throughout the study period. Similar trends were observed when the subgroups of high- and intermediate-risk PE were examined separately ([Figures 2B and 3B](#)). Temporal trends in the use of advanced reperfusion therapy, as well

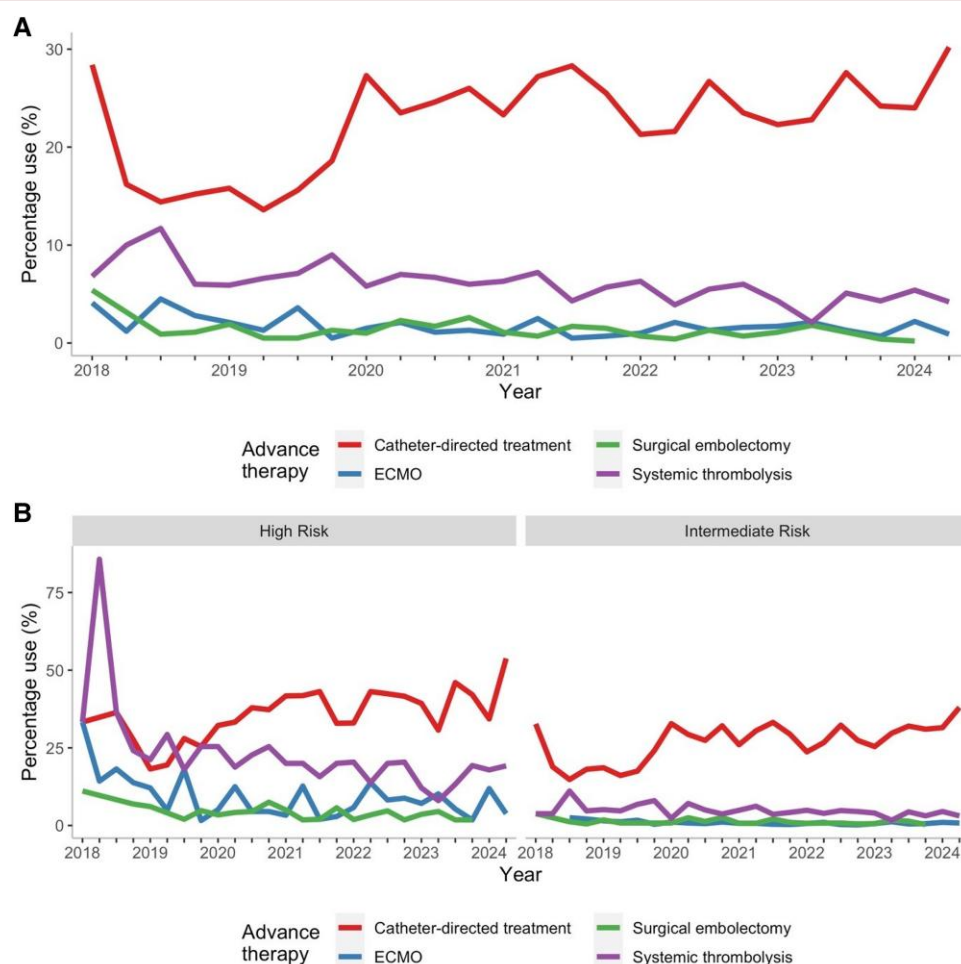


Figure 2 Time trends of various modalities of advanced therapy for pulmonary embolism in the PERTTM consortium registry, (A) in the entire patient population, and (B) in the high-risk vs. intermediate-risk category. ECMO: extracorporeal membrane oxygenation.

as specific CDT modalities, remained unchanged in a sensitivity analysis restricted to sites with data available for four or more years in the PERTTM Consortium registry ($n = 8265$ patients, 72.3% of the full cohort), confirming that observed temporal trends in catheter-directed treatment were not driven by changing site participation (data not shown).

Matched comparison of the PERTTM Consortium registry with the NIS

After propensity score matching, 3586 patients of the PERTTM Consortium registry were matched to 3586 hospitalizations of acute PE from the NIS across the period 2018–2020. Matched variables in the two groups of patients are shown in [Supplementary material online, Table S3](#). The Love plots with standardized mean differences for each matching variable are provided in the [Supplementary material online, Figures S1](#). The use of advanced therapies was lower in the NIS, especially regarding CDT, where it reached 1.6% in comparison to 20.7% in the PERTTM Consortium Registry. Similar differences were observed in the high-risk subgroup (2.6% vs. 29.2%).

Clinical outcomes in the matched comparison for the overall and the high-risk population are presented in [Table 2](#). Overall, in-hospital all-cause mortality was 6.2% in the PERTTM Consortium

registry vs. 6.5% in the NIS (between-group difference -0.3% , 95% CI -0.9% to 1.4%); the difference was significant and particularly pronounced in the high-risk population (12.4% vs. 31.3%; between-group difference -19% , 95% CI -24% to -14%). In-hospital major bleeding was observed in 4.4% of patients in the PERTTM Consortium registry vs. 9.1% in the NIS (between-group difference -4.7% , 95% CI -5.9% to -3.6%); again, the difference was largest in the high-risk population (-17% , 95% CI -22% to -12%). Length of stay was similar in the PERTTM Consortium registry with a median of 4 days (IQR 3–8) vs. 4 days (IQR 2–9) in the NIS (between-group difference -0.5 days, 95% CI -1.2 to 0.25), the difference was again significant and most pronounced in the high-risk population (-5.6 days, 95% CI -8.4 to -2.7).

In the exploratory secondary analysis, including patients from the overall available period of the two cohorts (2018–2024 for the PERTTM Consortium registry, 2016–2020 for the NIS database), in-hospital mortality rates differed between the two populations (between-group difference -1.3% , 95% CI -2% to -0.6%), and they remained significantly lower in the PERTTM Consortium registry when only the high-risk PE population was analysed ([Table 2](#)). In addition, length of stay was shorter in the PERTTM Consortium registry with a median of 4 days (IQR 2–8) vs. 5 days (IQR 2–9) in the NIS (between-group difference -0.75 days, 95% CI -1.3 to -0.2).

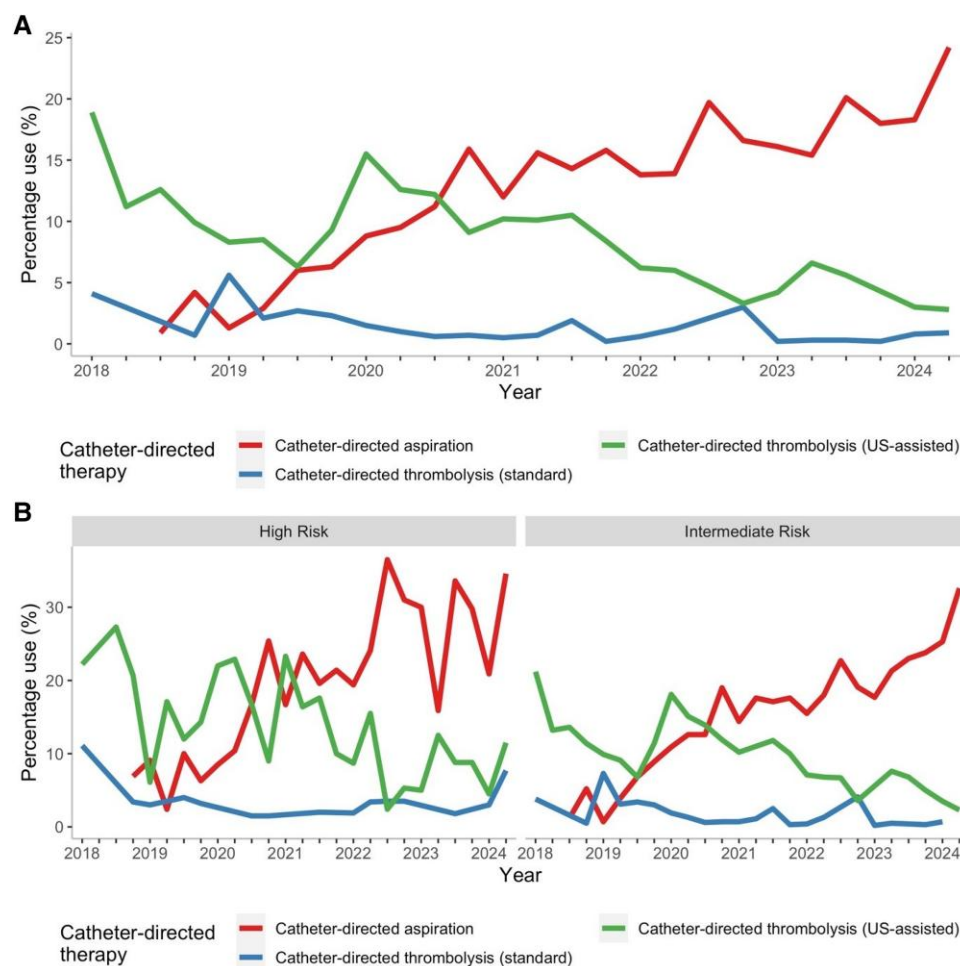


Figure 3 Time trends of various modalities of catheter-directed treatment for pulmonary embolism in the PERT™ consortium registry, (A) in the entire patient population, and (B) in the high-risk vs. intermediate-risk category.

Discussion

The present analysis from the PERT™ Consortium registry provides insights into the recent trends of treatment patterns and outcomes for patients with intermediate-high-risk and high-risk PE. CDT use increased over time, reaching 25.9% (and almost 40% in high-risk patients) in 2024, particularly as far as mechanical thrombectomy was concerned. Compared to the NIS, the PERT™ Consortium registry documented more frequent use of CDT and more favourable clinical outcomes, especially in the high-risk PE population. Keeping in mind that this is no direct comparison, our findings appear to support the hypothesis that recently introduced advanced therapies may be capable of improving the patients' prognosis.

Existing data on the use of advanced therapy, including trends over time, have mainly been derived from single-centre studies or nationwide administrative databases and referred to time periods ending before, or in the first year of, the COVID-19 pandemic. During that time, institutions implementing PERT recorded a substantial increase in CDT use, for example, from 1% in the period 2006–2012 to 14% in 2012–2016²⁰; in a recent systematic review of nine single-centre studies, the increase was more modest, from 3% to 6.4%.¹⁵ Data from the US Nationwide Readmissions Database revealed an overall increase in CDT use from 2.9% in 2016 to 7.1% in 2020²³; this trend was mainly driven by the increase in the use of mechanical thrombectomy.²⁴ In

Europe, annual rates of CDT have been very low during the same period,²⁵ but they are expected to increase substantially in the following years.¹⁹

Viewed against the background of the earlier reports mentioned above, the present analysis of the PERT quality assurance database registry reveals that the use of CDT continues to increase steadily, at least in some US hospitals, meanwhile reaching 27% for intermediate-risk and 37% for high-risk PE. Following a trend that began in the second half of the past decade, mechanical thrombectomy is more frequently performed compared to pharmacomechanical procedures in the US. This difference may be related to the potential of fast thrombus removal with large-bore devices, at least in experienced hands²⁶; by comparison, catheter-directed thrombolysis with or without ultrasound necessitates long infusion periods but is, on the other hand, easier to handle.²⁷ However, the clinical and prognostic implications of the conceptual and procedural differences between the two treatment modalities remain unclear. Uncontrolled data from registries have suggested favourable outcomes of patients treated with either method, and also with a more recently introduced system employing computer-assisted vacuum thrombectomy.^{26–28} An analysis of administrative data also reported no difference in clinical outcomes between the two techniques.²⁹ More recently, the PEERLESS randomized controlled trial compared large-bore mechanical thrombectomy with catheter-directed thrombolysis in 550 patients with intermediate-risk

Table 2 Comparison of in-hospital outcomes between the PERT™ consortium registry population and a matched cohort from the NIS

Patients included in the shared period 2018–2020					
All patients					
Characteristic	NIS n = 3586	PERT Consortium n = 3586	Difference ^a	95% CI	P value
In-hospital mortality (n, %)	231 (6.5%)	222 (6.2%)	0.26%	−0.89%, 1.4%	0.7
In-hospital major bleeding (n, %)	328 (9.1%)	158 (4.4%)	4.7%	3.6%, 5.9%	<0.001
Length of stay (days)	4 (2, 9)	4 (3, 8)	−0.50	−1.2, 0.25	0.2
High-risk patients					
Characteristic	NIS n = 469	PERT Consortium n = 469	Difference ^a	95% CI	P value
In-hospital mortality (n, %)	147 (31%)	58 (12%)	19%	14%, 24%	<0.001
In-hospital major bleeding (n, %)	119 (25%)	39 (8.3%)	17%	12%, 22%	<0.001
Length of stay (days)	12 (6, 23)	6 (4, 13)	5.6	2.7, 8.4	<0.001
Patients included in the overall period					
All patients					
Characteristic	NIS n = 10 883	PERT Consortium n = 10 883	Difference ^a	95% CI	P value
In-hospital mortality (n, %)	885 (8.2%)	749 (6.9%)	1.3%	0.56%, 2.0%	<0.001
In-hospital major bleeding (n, %)	1083 (10.0%)	370 (3.4%)	6.6%	5.9%, 7.2%	<0.001
Length of stay (days)	5 (2, 9)	4 (2, 8)	−0.75	−1.3, −0.20	0.008
High-risk patients					
Characteristic	NIS n = 1516	PERT Consortium n = 1516	Difference ^a	95% CI	P value
In-hospital mortality (n, %)	584 (39%)	170 (11%)	27%	24%, 30%	<0.001
In-hospital major bleeding (n, %)	355 (23%)	86 (5.7%)	18%	15%, 20%	<0.001
Length of stay (days)	11 (5, 21)	6 (3, 12)	5.2	3.4, 6.9	<0.001

^aStandardized mean difference, calculated by Welch two-sample t-test or two-sample test for equality of proportions as appropriate.
CI = Confidence interval.

PE. Patients receiving mechanical thrombectomy had shorter intensive care unit and total hospital stays as well as fewer all-cause readmissions, but there was no significant difference in mortality, intracranial haemorrhage, or major bleeding.³⁰ Thus, the data available so far add further support to the PERT concept, according to which decisions on the need for advanced therapy and the type of treatment to choose should be based on a holistic assessment of (i) the patient's clinical presentation, degree of haemodynamic instability and comorbidity profile; (ii) the expertise and resources locally available and (iii) the feasibility of referral to an expert large-volume centre.

In our study, patients in the PERT™ Consortium registry had lower in-hospital mortality rates and shorter length of hospital stay compared to the NIS matched cohort in the exploratory analysis of the overall populations, but not in the analysis focusing on the same inclusion periods of the two cohorts. However, mortality rates, major bleeding rates, and

length of stay were consistently lower in the high-risk population of the PERT™ Consortium registry. These findings support the hypothesis that CDT use guided by PERT consensus may improve patient outcomes and optimize efficiency of in-hospital treatment to an extent that can justify the increased costs of advanced reperfusion based on catheter interventions.¹⁹ A number of large randomized controlled trials are currently comparing CDT to the current standard of care with regard to clinically relevant endpoints, including long-term follow-up and patient-reported outcome measures.¹² Among them, the HI-PEITHO trial is testing ultrasound-assisted catheter-directed thrombolysis vs. anticoagulation alone in patients with enriched criteria of intermediate-high-risk PE.³¹ The PEERLESS II trial is comparing large-bore mechanical thrombectomy against anticoagulation in PE patients at intermediate risk.³² Furthermore, PE-TRACT, a trial funded by the National Heart, Lung, and Blood Institute of the US National

Institutes of Health, is primarily investigating whether CDT (all available modalities) will increase, compared with anticoagulation alone, the peak oxygen uptake on cardiopulmonary exercise testing at 3 months and reduce NYHA Class at 12 months.³³ In this context of ever-changing PE management landscape, efforts are underway to standardize data collection in PE care, addressing critical evidence gaps and enabling robust research to guide future treatment strategies.³⁴ These initiatives aim to harmonize management practices, improve outcomes, and build a stronger foundation for evidence-based innovations in PE care.

The present prospective report represents the largest and most contemporary multicentre analysis of CDT trends. However, some limitations and caveats should be noted when interpreting our results. First, the PERTTM Consortium registry includes only patients with at least one PERT consultation and, as such, patients with PE treated without the involvement of the site's PERT may have been missed, potentially introducing a selection bias in our results. However, the proportion of patients lost for this reason would be expected to be very low, since the participating sites are committed to involving their PERT in all patients with acute severe PE, and as such, the risk of bias is limited. Second, caution is warranted when extrapolating our findings to non-participating hospitals or countries outside the US, since the PERTTM Consortium registry involves a relatively high proportion of tertiary care centres with the experience and infrastructure for advanced treatment of PE. Third, the sites joined the registry at different times and contributed with varying numbers of patients over time. Accordingly, our analysis focused on patient-level characteristics without accounting for site-specific factors, preferences, or clustering in management or outcomes. However, a sensitivity analysis restricted to sites with data available for four or more years in the PERTTM Consortium registry confirmed the main results. Adjusting for hospital-level characteristics was not possible in this analysis, since the PERTTM Consortium registry does not capture these variables. Fourth, the PE risk class was entered in the analysis as captured in each patient's case report form and the registry database; no independent cross-checking of stratification based on individually captured baseline parameters was performed, except for patients with missing risk stratification by the site's PERT. Because of this, there may be some deviations from previously published reports from the PERTTM Consortium registry.¹⁸ In the intermediate risk strata, distinction between intermediate-low and intermediate-high risk was not possible. Finally, although the timeframes with the NIS partly overlapped, the PERTTM Consortium registry also included patients from more recent years (2018–2024) compared the NIS (2016–2020). We tried to overcome this limitation focusing on patients included in the overlapping time period (2018–2020) of the two databases and performing an exploratory analysis with the whole populations in order to ensure the consistency and comparability of the results.

In conclusion, our study highlights a significant and constant rise in the use of CDT, particularly mechanical thrombectomy, in a nationwide quality assurance database registry of PERT centres in the US. Management in the PERTTM Consortium registry was associated with lower in-hospital mortality, bleeding rates, and shorter hospital stay compared to nationwide practices, especially in the high-risk population, supporting the value of multidisciplinary care and advanced treatment implementation. The increasing use of CDT underscores the need for robust randomized trials to confirm its efficacy, safety, favourable effects on long-term outcomes, and overall cost-effectiveness in the evolving landscape of PE management.

Impact for daily practice

The increasing adoption of catheter-directed therapy (CDT), underscores a paradigm shift in the acute treatment of PE, driven by multidisciplinary pulmonary embolism response teams (PERTs). Clinicians may

consider early PERT activation to facilitate timely risk stratification and individualized therapeutic decisions, including the potential use of advanced reperfusion strategies. In-hospital outcomes, including mortality, major bleeding, and length of stay, differed between patients in the PERT registry and the NIS. Although descriptive, the results of the present study appear to support the potential value of multidisciplinary decision-making and access to catheter-directed treatment in PE care. As evidence continues to evolve, clinicians should remain updated on emerging trial results to guide optimal patient selection and procedural choice, balancing efficacy, safety, and resource availability.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

I.T.F. contributed to conceptualization, data curation, formal analysis, investigation, methodology, supervision, writing of the original draft, and writing—review and editing. S.H. contributed to data curation, formal analysis, investigation, methodology, supervision, and writing—review and editing. J.M.M., M.E., T.T., R.P.R., E.L., M.D.L., S.I.S., K.R., R.L., and E.S. contributed to investigation, project administration, supervision, and writing—review and editing. K.C.C. contributed to formal analysis and writing—review and editing. L.H. contributed to visualization, and writing—review and editing. L.V. contributed to formal analysis and writing—review and editing. S.B. contributed to conceptualization, supervision, validation, and writing—review and editing. S.V.K. contributed to conceptualization, funding acquisition, investigation, methodology, supervision, validation, visualization, writing of the original draft, and writing—review and editing.

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Data availability

The data from the PERTTM Consortium registry are the property of The PERT ConsortiumTM (<https://pertconsortium.org/contact/>), to which proposals for analyses and data access queries can be submitted. The data from the NIS are owned by the Healthcare Cost and Utilization Project (HCUP). Complete database availability information is included in the online HCUP Central Distributor Database Catalog, and data access queries can be submitted directly to the HCUP (<https://www.distributor.hcup-us.ahrq.gov/Databases.aspx>).

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